

Rahul Tax Advisory — Project Documentation

Full-stack marketing and client portal application for a boutique CPA/tax advisory firm.

Generated documentation for repo structure, architecture, features, and deployment flow.

Project Overview

This project is a full-stack application for Rahul Tax Advisory. It blends a React marketing site with a protected client portal, backed by a FastAPI API and MongoDB data storage.

- Public marketing site with service, industry, results, audit, contact, and booking pages.
- Client authentication, registration, and protected portal access.
- Capture leads, booking requests, audit intake, and documents from users.
- AI-powered chat assistant integrated via streaming backend API.

Frontend Architecture

The frontend is built with React using Create React App, React Router, Tailwind CSS, and several UI and form libraries.

- Routes defined in frontend/src/App.js.
- Protected client route for /portal using AuthProvider and useAuth.
- Pages include Home, About, Services, Industries, Results, Audit, Contact, Book, Login, Register, and Portal.
- Shared layout, chat widget, and toast notifications across the site.
- Axios-based API client for backend calls and form submissions.

Backend Architecture

The backend uses FastAPI and connects to MongoDB via motor. It provides auth, lead capture, booking, audit intake, document storage, and chat streaming endpoints.

- FastAPI server defined in backend/server.py.
- JWT authentication with cookies and bearer headers.
- User registration, login, logout, and current user endpoints.
- Admin-only endpoints for lists of leads, bookings, and audits.
- Document endpoints scoped to authenticated clients.
- AI chat endpoint streams responses using emergentintegrations and Anthropics model.

Key Backend Routes

- GET /api/ — health check
- POST /api/auth/register — create account and set auth cookie
- POST /api/auth/login — authenticate and set auth cookie
- POST /api/auth/logout — delete auth cookie

- GET /api/auth/me — fetch current authenticated user
- POST /api/leads — capture a new lead
- POST /api/bookings — create booking request
- POST /api/audits — submit audit intake
- POST /api/documents — add document record for user
- GET /api/documents — list documents for current user
- POST /api/chat/stream — AI chat streaming endpoint

Authentication and Data Flow

Users can register or login. Successful auth issues a JWT token stored as an HTTP-only cookie. Protected pages rely on the auth state to grant portal access.

- Registration stores users in MongoDB with hashed passwords.
- Passwords are hashed using bcrypt.
- JWT tokens are signed with a secret and validated for access.
- Admin role is seeded using ADMIN_EMAIL/ADMIN_PASSWORD environment variables.
- Protected portal content is only available to authenticated users.

AI Chat Feature

The AI chat is a streaming assistant that uses a system prompt tailored for tax advisory conversations. It persists user and assistant messages in MongoDB for history.

- Endpoint /api/chat/stream streams model responses as SSE.
- Uses emergentintegrations.llm.chat with Claude Sonnet model.
- System prompt positions the assistant as Rahul Tax Advisory tax support.
- Replies include booking and audit CTAs, and avoid binding tax advice.

Project Files and Structure

- frontend/ — React app source and UI components.
- backend/ — FastAPI server and API logic.
- frontend/package.json — frontend dependencies and scripts.
- backend/requirements.txt — backend Python dependencies.
- backend/server.py — main API implementation.
- frontend/public/cpa-firm-capacity-growth-playbook.pdf — existing playbook PDF.

Technology Stack

- Frontend: React, React Router, Tailwind CSS, Radix UI, Axios, React Hook Form.
- Backend: FastAPI, Motor/MongoDB, PyJWT, bcrypt, Pydantic.
- Deployment: CRA frontend build, FastAPI backend service, MongoDB database.
- AI: emergentintegrations with Anthropic Claude Sonnet model.

How to Run

Frontend: use yarn start or craco start in the frontend directory. Backend: run uvicorn backend.server:app --reload with required env vars.

Recommended Environment Variables

- MONGO_URL — MongoDB connection string.
- DB_NAME — database name.
- JWT_SECRET — secret key for JWT.
- ADMIN_EMAIL / ADMIN_PASSWORD — admin seeding credentials.
- EMERGENT_LLM_KEY — API key for the AI chat service.

Summary

This repository delivers a complete firm website and client engagement platform. It is designed to market services, capture leads, manage client authentication, and support communication through an AI-enhanced chat assistant.